

Toward Designing for Social Support Through Compassionate Interactions

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ABSTRACT

Prior research shows that compassion can promote social connection and well-being. Compassion is characterized by noticing another's suffering, connecting empathically, feeling concerned, and responding in a caring way. In this research, we engaged sixteen first-time parents in research activities to investigate how compassion is understood, perceived, and experienced by new parents to inform the design of potential technology solutions to foster support between new parenting partners. We posit that interventions facilitating compassionate interactions between new parents can improve support for new mothers, promote relationship satisfaction, and thus positively affect the overall wellness of new mothers during the transition to motherhood.

CCS CONCEPTS

• **Human-centered computing** → **HCI theory, concepts and models; Empirical studies in HCI.**

KEYWORDS

Compassion; Life Transitions; Mothers; Maternal Wellness; Compassion cultivation, parenthood, fatherhood, motherhood, support intervention; Journey Maps

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1 INTRODUCTION

Social support is commonly studied as a predictor of better mental and physical health during women's transition to motherhood [21]. Availability of adequate support has been associated with lower maternal postpartum depression and anxiety [1, 20, 22, 30]. Life transitions such as parenthood can alter relationships, existing patterns of interactions, expectations of support, routines, roles, and identities. In some cases, the individuals going through the transition may be surrounded by people; still, they can feel isolated and unsupported. Often, the changes associated with the transitions are not understood by the person going through transition and their

supporters, resulting in miscommunications and misunderstandings. They can create a gap between expected and received support, leading to disappointments, isolation, relationship breakdown, and thus additional stress. Indeed, previous research has identified a gap in mothers' expectations of help and received help. [13, 26].

As we explored past research to identify potential pathways to improved support to reduce the identified support gap, compassion emerged as a potential mediator, as it promotes better relationships and well-being [6, 12, 14, 16]. Compassion is characterized by noticing another's suffering, connecting through empathically, feeling concerned, engaging in cognitive perspective-taking, and responding in a caring way [8]. Therefore, we posit that interventions that promote compassionate interactions can mediate between social support and well-being.

To design interventions that facilitate improved support during life transitions by fostering compassion, it is essential to develop a contextual understanding of the elements of compassionate interactions. Compassionate care is defined by the people who give and receive it, and therefore interpersonal processes that capture what it means to people are an important element of its promotion [7]. Therefore, in this study, we included new mothers and their respective partners in an effort to understand how the receiver (mother) and the giver (partner), in this context, define and perceive compassionate care. This study aimed to identify the elements of compassionate interactions and understand how they manifest through interactions in the context of a significant transition, transition to motherhood.

2 RELATED WORK

In this section we discuss the prior research that informed our study.

2.1 Social Support and Maternal Wellness

As women transition to motherhood, they need and expect additional support and compassionate care. According to Dewar, compassionate care is a relational activity that is about the way in which we relate to other human beings when they are vulnerable. It can be nurtured and supported. It involves noticing another person's vulnerability, experiencing an emotional reaction to this, and acting in some way with the person, in a way that is meaningful [7].

During transition to motherhood, support from a variety of sources helps safeguard mothers' physical and mental well-being [20, 21]. Though new mothers expect support from various sources, they consider partners as the main source of support [26, 36]. Research shows that enacted or received support from a partner surrounding pregnancy predicts maternal prenatal and postpartum mental health. However, previous studies have identified a gap in expected

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and received support from supporters including partners [13, 26, 27]. A longitudinal study of pregnancy found that effective prenatal support from the baby's father was associated with relationship satisfaction and interpersonal orientation. It also predicted decreases in prenatal anxiety across pregnancy [29]. Informed by the important role new mothers' partners play in overall wellness and relationship satisfaction, in this study, we focus on compassionate interactions between the mother and her partner.

2.2 Compassion and Wellness

The notion of compassion is universally understood yet simultaneously hard to define and communicate [35]. The essential qualities of compassion are motivation to be caring, sensitive, and moved by distress. Chochinov defines compassion as a deep awareness of the suffering of another coupled with a desire to alleviate it [9, 10]. Compassion differs from both empathy, which is experiencing others' feelings, and sympathy, which is a feeling of sorrow and concern for others' pain or suffering [15]. Compassion is conceived of as a feeling of concern for another person's suffering (empathy) **accompanied by a motivation to help**.

Researchers from different backgrounds have studied compassion and found that compassion is positively associated with life-satisfaction, happiness, optimism, and social connectedness, as well as inversely associated with depression, and anxiety [4, 16, 23, 24]. A plethora of previous research has shown that developing care and compassion for self and others has benefits ranging from psychological processes to social relationships [5, 25, 33]. There is also growing evidence that social connection and compassion are important predictors of health and well-being [17, 18, 28, 32, 34]. The impetus for this research is inspired by the potential of technology-mediated compassion cultivation to improve the overall wellness of individuals transitioning to parenthood.

There are a few compassion-related interventions designed by HCI researchers. Lee et.al. created a self-compassion chatbot (Vincent) and compared between caregiving and care-receiving conditions. They found that while self-compassion increased for both conditions, only those with care-receiving Vincent significantly improved [19]. Schrie et.al. piloted game jam to enhance perspective-taking, empathy, and compassion across different ethnic groups from public schools in Nigeria [31].

3 METHOD

In this section, we provide details of recruitment, study participants, and research activities. The study included semi-structured interviews and a two-week-long journey-mapping activity.

3.1 Participants

We recruited participants by posting recruitment messages in a local parenting group. Inclusion criteria included 1) first-time parents with a child less than one year old, 2) the availability of both parents to participate in the study, and 3) both parents living in the same household. We were able to recruit eight couples (8 mothers, and 8 fathers) who were first-time parents. Even though it was not intentional, all the participants were white, married, and in heterosexual relationships. Their age ranged from 23-26 years. Table 1 shows the education and employment information of the participants.

	Age	Education	Employment
Mom1 (M1)	23	Bachelor	Part-time
Mom2 (M2)	24	Bachelor	Homemaker
Mom3 (M3)	23	Bachelor	Full-time
Mom4 (M4)	25	Masters	Full-time
Mom5 (M5)	23	Some College	Homemaker
Mom6 (M6)	24	Bachelor	Homemaker
Mom7 (M7)	25	Professional	Full-Time
Mom8 (M8)	24	Some College	Homemaker
Partner1 (P1)	24	Bachelor	Masters Student
Partner2 (P2)	25	Bachelor	Full-time
Partner3 (P3)	24	Bachelor	Full-time
Partner4 (P4)	26	Masters	Doctoral Student
Partner5 (P5)	24	Bachelor	Full-time
Partner6 (P6)	25	Masters	Doctoral Student
Partner7 (P7)	26	Bachelor	Full-time
Partner8 (P8)	24	Bachelor	Full-time

Table 1: Participants - Interviews and Journey Mapping

3.2 Study Activities

3.2.1 Semi-Structured Interview. The primary purpose of the interview was to understand the parenting partners' perceptions of compassion. The interview focused on couples' daily interaction experiences. The interview schedule was loosely divided into the following areas: the nature of the interaction between the couple, information about positive and negative daily interactions, communication methods and channels, perceptions of elements of compassionate interactions, participants' definitions of compassion, and barriers to compassionate interactions.

The interviews lasted for 25 - 45 minutes. The interviews were audio-recorded and later transcribed for analysis. Participants signed the informed consent in person prior to the start of the interview.

3.2.2 Journey Mapping. We developed customized journey maps for mothers and partners. At the end of the semi-structured interview, the couples were introduced to the journey mapping activity. We gave them file folders with fourteen journey map pages for two weeks of journey mapping activity (one for each day). They were also instructed to write daily reflections. A sample of the completed journey map by a mom participant is included in Appendix. Each participant was paid \$50 for their participation in the research.

4 ANALYSIS

Guided by grounded theory [2, 11], we thematically analyzed the transcribed interviews and journey map entries. We coded the collected data to identify recurring themes and patterns and used mind mapping as a theme visualization tool [3].

5 FINDINGS

In quotations we use in this section, Mom participants are denoted as "M1-M8" and partner participants as "P1-P8". The quotes from the journey mapping activity are denoted by "JM" along with the section names in the journey map (feeling, doing, thinking, and reflection). Other quotes are from the interviews.

5.1 Communicating the Needs and Frustrations

Becoming a mother can be both exciting and terrifying. New parents want what's best for the baby, but as first-timers, the mother and her partner learn as they go. Our study participants experienced

difficulty communicating each other's needs and feelings during the transition to new parenthood.

P7: *"Looking back something that would have been helpful through that transition, it would have been helpful if we had established beforehand things we can do to communicate a little more efficiently. If she's not feeling well or she's upset or having mood swings, maybe finding a way to communicate that better so I can understand how to help her a little more."*

M6 (JM:Thinking): *"I think I didn't realize how frustrated and upset I was until we had a good talk."*

5.2 Noticing the Struggles

According to mothers in our study, compassion involves noticing their struggles. They consider their partner noticing and acknowledging their hardship and proactively helping them as compassionate.

M7: *"For me it's I think without asking for help and just- I don't know just people recognizing that I'm having a hard time and be like hey let me help you with this."* Participant M5 conveyed a similar feeling in the interview.

M5: *"If I feel like I'm the one doing everything, I start to get overwhelmed. I wish that he would see some of my needs more. Or if I notice that I'm asking him to help me a lot that's kind of the same thing. I feel like I just wish he would notice."*

5.3 Understanding

Our findings show that the lack of understanding of the needs of new mothers was a barrier to compassionate interactions. This situation is especially challenging since new mothers themselves were unaware of the demands of caring for an infant during the transition to motherhood.

P7: *"There were times, particularly when she was pregnant, where I had trouble understanding what her mood was every once in a while. Of course, that was something that either I couldn't tell what she was feeling or it was unclear to me."*

M5: *He doesn't understand my career, keeps expecting me to come home too early, etc.*

5.4 Listening

According to our participants, being compassionate involved a willingness to listen to hopes, frustrations, and fears.

P3: *"If I feel like we need it, I'll listen and let her vent or let her talk about whatever she wants to talk about and kind of let her get it out. Other times I just give her space. I think there's different types of frustration."*

A willingness to listen encourages mothers to talk about frustrations and stresses and thus helps convey their needs.

M8 (JM: Feeling): *"It made me feel a lot more relaxed that he [P8] was willing to sit and listen to my concerns and fears, and honestly talk about our options."*

5.5 Feeling of Being Cared for

Caring involves displaying kindness and concern for others and providing help when needed. Our participants shared instances where their partners showed concerns for them, which made them feel cared for.

M6: *"He texted to ask about my day and showed empathy when I indicated that I had a difficult drop off with my son."*

M2: *"My husband brought the baby down to cuddle with us in bed, something we only do on weekends. It was a special and great way to start the day."*

5.6 Collaboration In Parenting and Social Activities

The feeling of partnership and collaboration helped mothers feel good about themselves and their relationships.

M4 (JM: Reflection): *"He [P4] is all in on parenthood and marriage, and it makes me feel special and my heart full."*

When there is a lack of collaboration and partnership in parenting responsibilities and in many cases, in social activities, new mothers feel a lack of care and compassion.

M3: *"There are plenty of times when my husband schedules work-related dinners or one of his many hobbies (fishing, hunting, wood-working) without asking me if I am able to stay with the kids. There seems to be an assumption in our household that I am always available for the kids."*

5.7 Planning Ahead to Avoid Resentment

Lack of planning and uncertainty frustrated the mothers in our study.

M1: *"...when you're in the moment it's really easy to resent them for not doing what you want them to do, but if you have a plan like alright if he[their son] wakes up during this time window, like timeframe, you go. If it's this time-frame I'll go, so to be better prepared and have specific roles."*

The couples who had established their expectations, and schedules seemed to have better and happier relationships.

5.8 Sharing to Lessen the Burden

Our participants, especially the moms, shared that they felt happy when they worked together with their partners.

P4 (JM: Doing): *"We worked together to keep him [our son] entertained. The feeling of teamwork was nice."*

P8 (JM:Thinking): *It's nice to be able to share how I feel without feeling like an extra burden on her day.*

5.9 Awareness of the Needs without Prompting

Mothers expected their partners to be aware of their needs. Sometimes the help partners offer differs from what the mothers need or prefer. A journey map entry by M4 for one weekend evening indicated that she was irritated and annoyed with the help she received. She said,

M4: *"My husband's been baking bread all day, and I wish he'd take a more active role with the baby."*

One would think that the mother is being unappreciative of her husband, but she explained the reason for her irritation with the accompanying "thinking" response recorded in the journey map simultaneously.

M4: *"I look forward to a caretaking break on weekends and I don't have one."*

She longed for a break from caring for the baby during the weekend. Instead, she took care of the baby while her husband baked.

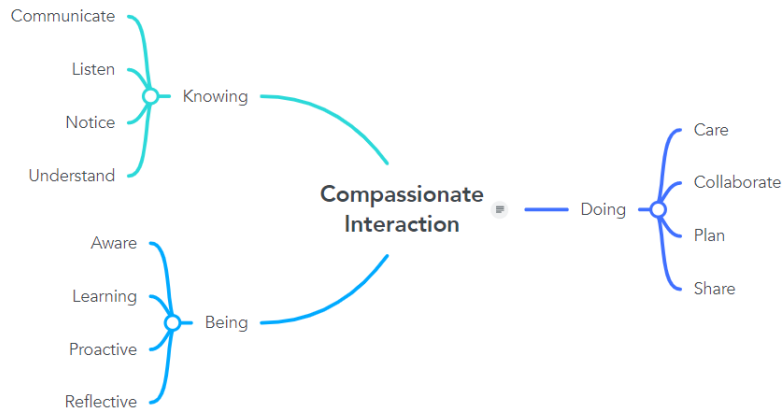


Figure 1: Elements of Compassionate Interaction as a Mindmap [3]

5.10 Continuous Learning

According to our participants, they had to continuously cope with their own and their babies' evolving needs, whether recovering from childbirth or caring for a newborn.

M5: "I think I wish I knew that the physical recovery from having the baby would be difficult. I wish I knew how hard the lack of sleep was going to be. I wish I knew more about the importance of having a schedule for a baby."

M1: "For the first few weeks and even months it was just me and us kind of figuring out how we're gonna just deal with the transition of having a baby, but definitely once he got older and I expressed I need some time alone, I need you to watch the baby or I would love to be able to sleep in a little bit because I woke up five times last night. It definitely took some time for us to figure that out."

5.11 Proactive Support

Previous studies showed that new mothers are reluctant to ask for help. Mothers in the study expressed frustration when they had to ask for help. They shared that they feel compassion when their partner is proactive in helping them.

M2: "I feel compassion from him for me when he does things without me asking."

Some participants shared that being proactive helped reduce their stress.

M6 (JM:Doing): "We had friends over for dinner, so I got some social time, then Taylor and I figured out our week and worked on a project together after Ellie went to bed. It makes it feel like not everything is so stressful."

5.12 Reflective

Study participants shared that reflecting on their thoughts and feelings and the realities of day-to-day experiences help them to learn about the needs of each other.

M6 (JM: Thinking): "Talking about what is on our hearts/minds is really good for our relationship."

M8 (JM: Thinking): "Got on the same page with husband, both emotionally and schedule."

6 DISCUSSION

The findings from the study revealed the evolving nature of the support needs and expectations and the potential for compassionate interactions to bridge the gap in expected and received support identified in the previous studies. We rely on previous research to present the elements of compassionate interactions that emerged from the data analysis cohesively and clearly. Past research conceptualized compassion as a three-part activity composed of (1) noticing another person's suffering; (2) connecting with them; and (3) responding to the suffering. For example, Way et al. described the compassionate communication of hospice workers as a three-part activity of recognizing, relating, and (re)acting [37]. We extended this prior work by conceptualizing compassionate interactions during life transitions as a three-part interactive process of "**knowing**," "**doing**," and "**being**" and classified the identified elements (from section "findings") into those three parts. This classification is depicted as a mindmap in Figure 1.

We posit that during stressful life transitions such as the transition to motherhood, the individuals going through transitions, as well as their supporters go through cycles of "knowing" (how to be compassionate), "doing" (interacting compassionately), and eventually "being" compassionate partners.

6.1 Visualizing Compassionate Interactions - The Compassion Wheel

During life transitions, compassion is not just a single event or a one-time activity. It is an ongoing process, a dynamic set of interactions, continuously evolving along the transitional phases. The compassion process is not disconnected. It is not a static process either. During the different phases of transitions, the process of "knowing," "doing," and "being" needs to be re-enacted. For example pregnant mother and the partner need to "**know**" about the

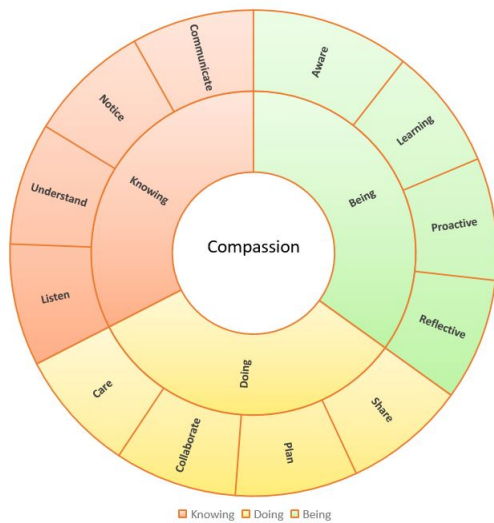


Figure 2: The Compassion Wheel - A conceptual Model of Elements of Compassionate Interactions During Life Transitions.

challenges, changes, and needs by communicating, noticing, understanding, and listening. Informed by this knowledge, they need to "do" care, collaborate, plan, and share. Hopefully, they can "be" compassionate by being aware, proactive, reflective, and constantly learning. Then, after the childbirth, the process of knowing, doing, and being under the new challenges and needs resumes, and the cycle continues. To signify the ongoing and continuous nature of compassionate interactions during life transitions, we present the conceptual model of compassionate interaction in the shape of a wheel we call the Compassion Wheel (Figure 2).

7 IMPLICATIONS FOR DESIGN

The identification of the elements of compassionate interactions helps designers to conceptualize potential social support interventions that facilitate and trigger compassionate interactions and expose and communicate support needs resulting in improved wellness. This idea is captured in figure 3. In the specific context of

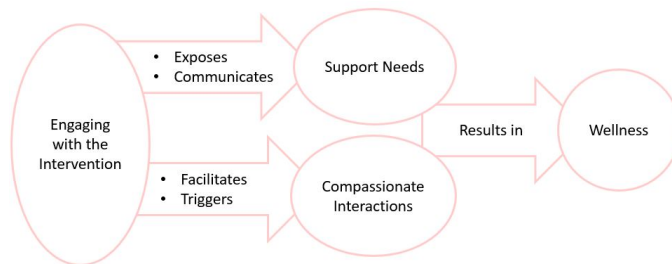


Figure 3: Conceptualizing Compassion Cultivating Intervention

transition to parenthood, potential compassion-cultivating interaction could: 1) expose and communicate evolving needs of new

mothers and partners ("knowing"), and 2) promote and facilitates compassionate interactions between the partners ("doing") resulting in reduced support gap and thus improved wellness ("being"). This concept can be applied to other life transitions.

In addition, interaction designers can use, adapt, or extend this research to investigate elements of compassionate interactions for different contexts of life transitions, be it disease diagnosis, divorce, traumatic experiences, or rehabilitation.

8 LIMITATIONS

The main limitations of this study are the small number (16) and the lack of diversity of participants. These may limit the generalizability of the results. However, we hope future research will expand the population and demographic reach in this design space.

9 CONCLUSION AND FUTURE WORK

This research identified an unexamined and novel design space - designing for compassionate interactions during life transitions - to facilitate better support. We investigated a significant life transition in society - transition to parenthood - to uncover elements of compassionate interactions. Insights from the investigation helped us conceptualize the elements of compassionate interactions during life transitions as a continuous process of knowing, doing, and being. To signify the ongoing and continuous nature of compassionate interactions during different phases of life transitions, we developed the compassion wheel - a conceptual model of the elements of compassionate interactions during life transitions.

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A JOURNEY MAP

A.1 A completed Journey Map

Daily Reflections: (Time: 6 pm)

Please reflect on the interactions you had with your partner today. You may reflect on the following prompts. You may reflect on any other topics you wish to share.

What are your thoughts and feelings about your interactions with your partner today?

What interactions of him made you happy? What made you unhappy?

(If relevant): What interactions of you made him happy? What made him unhappy?

Positive interactions today.
My husband made me happy by helping with housework and helping with my migraines.
I made him happy by sitting aside time to sit together.

Mother's Journey Map

Participant ID: _____

	Morning (Time: <u>8:00 am</u>) (Reflect on time between previous recording and now)	Evening (Time: <u>2:00 pm</u>) (Reflect on time between previous recording and now)	Before Sleep (Time: <u>9:30 pm</u>) (Reflect on time between previous recording and now)
Feeling	 Exhausted, frustrated. Not a good night at work. lots of stress.	 Rested, I was able to sleep.	 Content, Happy
Doing	Who did you interact with recently/ now? What was the interaction? Did that contribute to how you feel? Co-workers, patients/families, my family. I am still adjusting to a new job, and I still feel isolated from the other staff members. So, when it is stressful I don't feel supported.	Who did you interact with recently/ now? What was the interaction? Did that contribute to how you feel? Just family (husband/baby). Robert watched the baby so that I could sleep, and they took him on an errand, so I have had about an hour of alone time, which I really needed today.	Who did you interact with recently/ now? What was the interaction? Did that contribute to how you feel? We had Thanksgiving dinner with some friends, and we skipped with my in-laws. It really helped to have some interaction and to talk to people that love us.
Thinking	What are your thoughts now about how you feel? Work is hard. Working with a baby is hard. Working nightshift with a baby and a husband in grad school in a state on the opposite end of the country from your family is hard. I am so tired.	What are your thoughts now about how you feel? I'm glad I've had some alone time to recharge after my terrible night at work. I wish I had more support at my job (including pumping time).	What are your thoughts now about how you feel? I miss my parents a lot. I am glad we have a network of friends here to support us, but it isn't always the same as having family.

Figure 4: A completed Mother's Journey Map Entry for a Day